CPE 314 — Software Design

University of Cebu Computer Engineering Department

FleetRent: Vehicle Rental & Fleet Management System

Project Design Document

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Revision History

Table 1: Revision history of this document

Version	Date	Author	Changes
0.1	October 21, 2026	Gabasa - One Paragraph Pitch Coloscos - ER Sketch Labra - Feature Checklist Rosal - Task split Lagaya – Public API and Risks/Open questions	Initial draft — Week 8 sections 1 to 6
0.2	October 27, 2026	Coloscos - Section 3 (Class Hierarchy and SRP) Gabasa - Section 4 (ERD and Schema)	Week 9 sections added — Section 3 and Section 4
0.3	November 4, 2026	Labra - Section 5 (API) Lagaya - Section 6 (Concurrency) Coloscos - Section 7 (Data Mining & Visualization) Rosal - Section 8 (Testing Summary)	Week 10 sections added — Sections 5, 6, 7 and 8
1.0			Final compilation and polish — Week 11

1. Introduction & Project Overview

FleetRent is a JavaFX desktop app that helps small shops renting out cars, SUVs, and motorcycles replace their paper logbooks, which make it easy to lose track of which vehicle is out, its mileage, and who is next in line. It stores vehicles, renters, and rentals together in a single SQLite database, so staff can check vehicles out and in, look up rental history by renter or vehicle, and undo the last check-out if they make a mistake. When a vehicle is already rented, the app keeps a waiting list so the next customer is served in order, and it pulls verified make and model data from the NHTSA vPIC API instead of relying on typed-in entries. A dashboard then shows the most rented vehicle type and how busy the fleet is over time, giving owners a clear, organized view of their business.

2. Preliminary ER Sketch & Team Roles

This is the preliminary ER sketch from our Week 8 project proposal. It has three entities (Renter, Vehicle, and Rental) and two relationships. Tables 2 to 4 list the attributes of each entity, and Figure 1 shows how they connect. It is a rough draft and will be replaced by the finalized ERD in Section 4.

Table 2: Renter Entity (Preliminary Attributes)

Attribute	Key	Description
`id`	PK	Unique number for each renter.
`name`	-	Full name of the renter.
`contact`	-	Phone number of the renter.
`email`	-	Email address of the renter.
`address`	-	Home address of the renter.

Table 3: Vehicle Entity (Preliminary Attributes)

Attribute	Key	Description
`id`	PK	Unique number for each vehicle.
`type`	-	SEDAN, SUV, or MOTORCYCLE.
`make`	-	Brand of the vehicle (from the NHTSA vPIC API).
`model`	-	Model name of the vehicle (from the NHTSA vPIC API).
`year`	-	Model year of the vehicle.
`plateNumber`	-	License plate number of the vehicle.
`color`	-	Color of the vehicle.

Attribute	Key	Description
`dailyRate`	-	Rental price per day.
`mileage`	-	Total miles the vehicle has driven so far.
`lastPMS`	-	Date of the last preventive maintenance service (PMS).
`categoryPath`	-	Category text such as `Car > Sedan > Compact Sedan`, used to build the category tree.

Table 4: Rental Entity (Preliminary Attributes)

Attribute	Key	Description
`id`	PK	Unique number for each rental.
`renterId`	FK	Links to `Renter.id` (who rented).
`vehicleId`	FK	Links to `Vehicle.id` (what was rented).
`startDate`	-	Day the rental starts (rent from).
`endDate`	-	Day the vehicle is due back (rent to).
`returnedOn`	-	Day the vehicle was actually returned to the rental shop. Empty until it is returned.
`milesDriven`	-	Miles driven during this rental (mileage used).
`status`	-	`ONGOING` or `RETURNED`. If the vehicle comes back earlier than `endDate`, the status becomes `RETURNED` and the vehicle is available to rent again.

Relationships

One Renter can have many Rentals (1:M), and one Vehicle can be in many Rentals over time (1:M). The Rental table connects a renter to a vehicle.

The Queue (waiting list) and the Stack (undo) are kept in memory only, so they are not tables. `categoryPath` is a text such as `Car > Sedan > Compact Sedan` that we use to build the vehicle category tree in code.

When a vehicle is returned, even earlier than its `endDate`, the rental status becomes `RETURNED` and the vehicle is available to rent again.

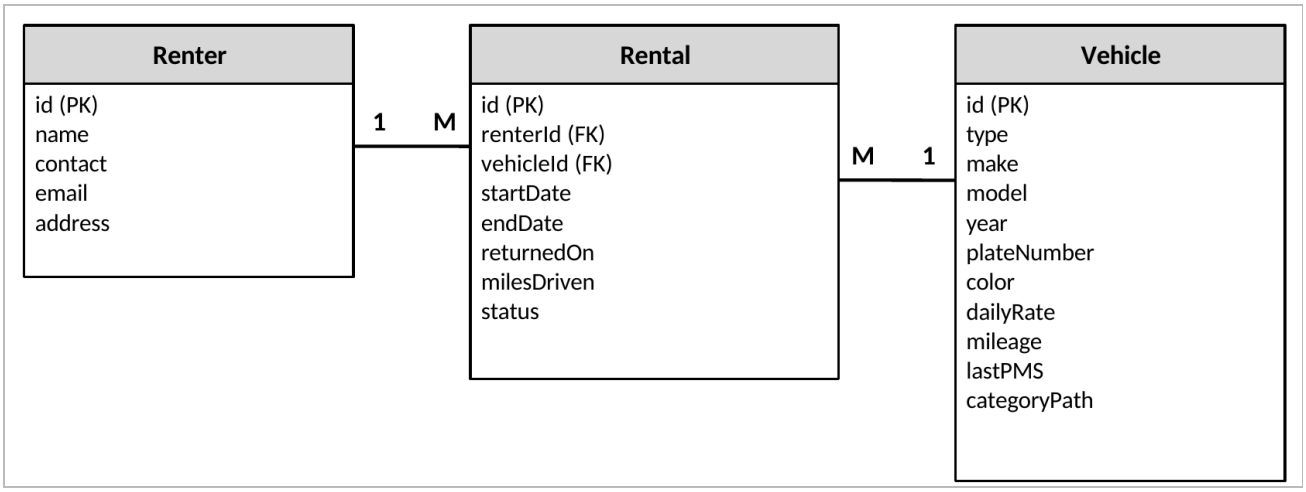


Figure 1: Preliminary ER sketch of FleetRent (Renter, Rental, Vehicle)

Team Roster and Primary Roles

Each member has one role and one main job for Milestone 1, the same as in our project proposal.

Table 5: Team Roster and Primary Roles (Group 4)

Member (GitHub Username)	Role	Primary Responsibility
Gabasa, Dexter (`gabasad84-afk`)	DB	Database: SQLite tables for vehicles, renters, and rentals, and the JDBC code with `PreparedStatement`. Database tests.
Rosal, Wendil (`wendilrosalx22`)	ADMIN, QA	Vehicle category tree and the recursive fleet mileage method. Maven project setup. Tests for these.
Coloscos, Brent (`Coloscos Brent`)	TEAM LEAD, DEVELOPER	Domain model and OOD: abstract `Vehicle` class, its three subclasses, and the rate and fee logic. Tests for these.
Lagaya, John Carlo (`caloylagaya2k4-pixel`)	DEVELOPER	Rental and Renter classes, the waiting list Queue, the undo Stack, and the CSV import on a background thread. Tests for these.
Labra, Simone (`simonelabra21`)	UI	NHTSA vPIC API: sending requests and parsing JSON. JavaFX GUI (forms, buttons, dashboard) and the bar and line charts.
All members	-	Commit our own work to the team repo, review each other's pull requests, and attend the Week 8–9 consultation.

3. System Architecture & Object-Oriented Design

FleetRent is split into four parts: the model classes (vehicles, renters, rentals), the in-memory data structures (waiting list, undo stack, category tree), the Database class that talks to SQLite, and the JavaFX window. The window never writes SQL, and the Database class never touches the window.

3.1 Class Hierarchy

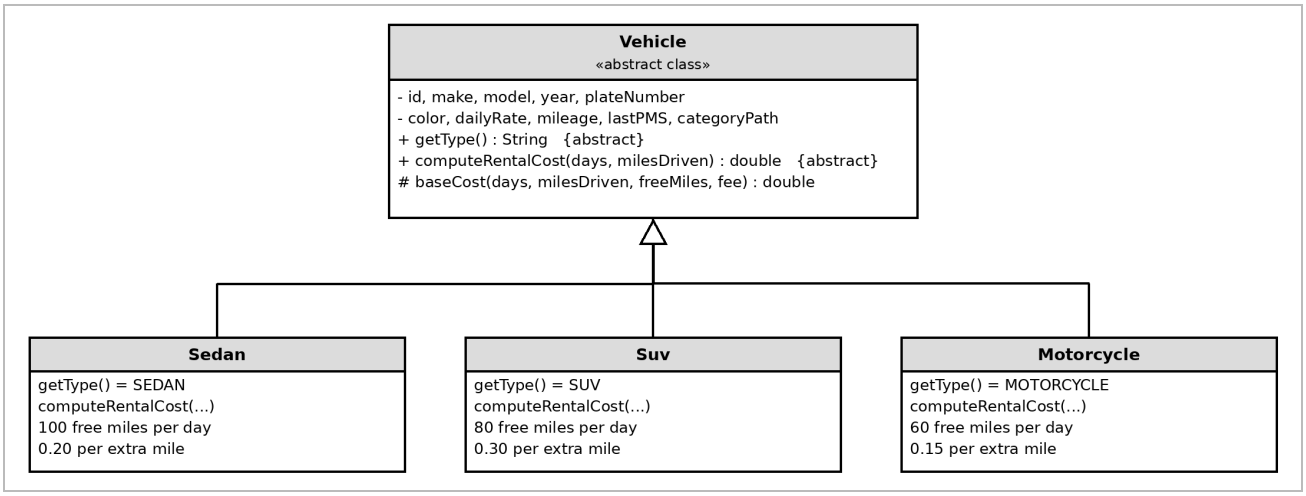


Figure 2: Class hierarchy of FleetRent (Vehicle and its three subclasses)

`Vehicle` is an abstract class. It holds the data every vehicle has (make, model, year, plate number, color, daily rate, mileage, last PMS date, category path) and has two abstract methods: `getType()` and `computeRentalCost(days, milesDriven)`. `Sedan`, `Suv`, and `Motorcycle` extend it. Each one answers `getType()` with its own type and has its own mileage fee rule (Table 6):

Table 6: Mileage Fee Rule of Each Vehicle Subclass

Subclass	Type text	Free miles per day	Fee per extra mile
Sedan	SEDAN	100	0.20
Suv	SUV	80	0.30
Motorcycle	MOTORCYCLE	60	0.15

These numbers are our current rules and can still change.

The hierarchy is used polymorphically while the app runs. The window and the database code hold plain `Vehicle` references. When the app calls `computeRentalCost(...)` on one, Java runs the `Sedan`, `Suv`, or `Motorcycle` version at runtime. `VehicleFactory` creates the right subclass from the type text stored in the database.

The other classes are not part of this hierarchy:

- `CategoryNode`: one box in the category tree (Car > Sedan > Compact Sedan). Its recursive `totalMileage()` adds up the mileage of a box and everything below it.
- `Renter`, `Rental`, `RentalStatus`: the data for a customer and a rental (ONGOING or RETURNED).
- `WaitingList`: a Queue of renters waiting for a vehicle (first in, first out).
- `CheckoutHistory`: a Stack of check-outs so the last one can be undone (last in, first out).
- `Database`: all SQLite code, using JDBC `PreparedStatement`.

- `App` and `Main`: the JavaFX window and the class that starts it.

3.2 Single Responsibility Principle

Table 7 explains, for each class, its one responsibility and why that follows the Single Responsibility Principle (SRP): each class has one reason to change.

Table 7: SRP Justification for Each Class

Class	Its one responsibility	Why this follows SRP
Vehicle	Holds the shared vehicle data and defines what every vehicle must be able to do	It does not save to the database or draw anything. It only describes a vehicle.
Sedan	The rental cost rule of a sedan	Only a change in the sedan fee rule changes this class.
Suv	The rental cost rule of an SUV	Only a change in the SUV fee rule changes this class.
Motorcycle	The rental cost rule of a motorcycle	Only a change in the motorcycle fee rule changes this class.
VehicleFactory	Creating the correct Vehicle subclass from a type text	Adding a new vehicle type changes this class, and no other class has to know how objects are created.
CategoryNode	The category tree and the recursive mileage total	It knows about categories and mileage only, not about rentals or the database.
Renter	The data of one renter	It only stores renter information.
Rental	The data of one rental and marking it as returned	It does not decide prices and does not save itself.
RentalStatus	The allowed rental states	It only lists ONGOING and RETURNED.
WaitingList	The waiting line of renters	It only manages the queue order.
CheckoutHistory	The undo stack of check-outs	It only manages the undo order.
Database	Reading and writing SQLite	It is the only class that contains SQL. If the database changes, only this class changes.
App	The JavaFX window and button handlers	It shows data and passes user actions on. It does not contain SQL or cost rules.
Main	Starting the JavaFX application	It only launches `App`.

4. Data Model & Database Design

4.1 Finalized ERD

This is the finalized ERD of FleetRent. It replaces the preliminary sketch from Week 8. It has the same three entities, now with data types. The labels 1 and M show the cardinality: one Renter to many Rentals, and one Vehicle to many Rentals. One Renter can have many Rentals (1 to M), and one Vehicle can be in many Rentals over time (1 to M). The Rental table connects a renter to a vehicle through two foreign keys.

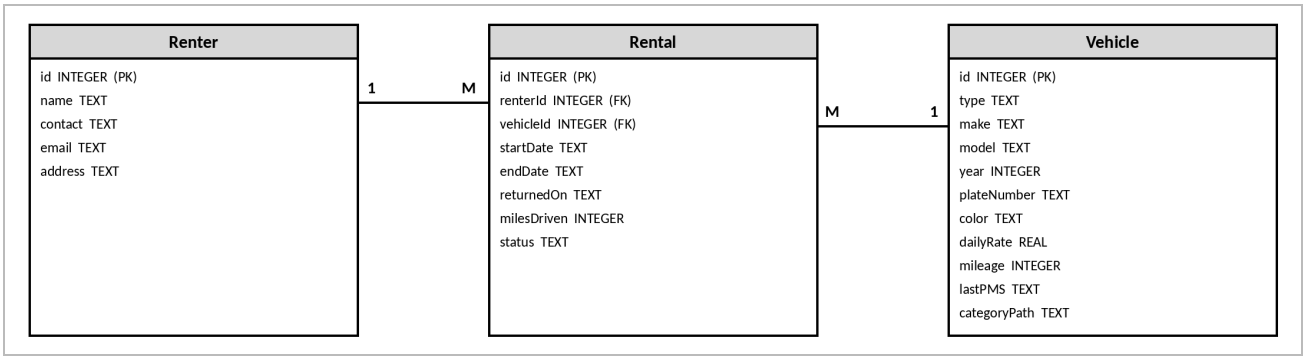


Figure 3: Finalized ERD of FleetRent (Renter, Rental, Vehicle)

The waiting list Queue and the undo Stack live in memory only, so they are not tables. The category tree is built in code from the `categoryPath` text of each vehicle.

4.2 Schema Description

The SQLite database file is `fleetrent.db`. The app creates it and the three tables the first time it runs, using the CREATE TABLE statements below (they are also saved in `src/main/resources/schema.sql`). They match the finalized ERD in Figure 3.

```
CREATE TABLE IF NOT EXISTS Renter (
    id       INTEGER PRIMARY KEY AUTOINCREMENT,
    name     TEXT NOT NULL,
    contact  TEXT,
    email    TEXT,
    address  TEXT
);

CREATE TABLE IF NOT EXISTS Vehicle (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    type          TEXT NOT NULL CHECK (type IN ('SEDAN', 'SUV', 'MOTORCYCLE')),
    make          TEXT NOT NULL,
    model         TEXT NOT NULL,
    year          INTEGER,
    plateNumber   TEXT,
    color         TEXT,
    dailyRate     REAL NOT NULL,
    mileage       INTEGER NOT NULL DEFAULT 0,
    lastPMS       TEXT,
    categoryPath  TEXT
);

CREATE TABLE IF NOT EXISTS Rental (
    id           INTEGER PRIMARY KEY AUTOINCREMENT,
    renterId     INTEGER NOT NULL REFERENCES Renter(id),
    vehicleId    INTEGER NOT NULL REFERENCES Vehicle(id),
    startDate    TEXT NOT NULL,
    endDate      TEXT NOT NULL,
    returnedOn   TEXT,
    milesDriven  INTEGER NOT NULL DEFAULT 0,
    status       TEXT NOT NULL DEFAULT 'ONGOING' CHECK (status IN ('ONGOING', 'RETURNED'))
);
```

Notes on the schema:

- Dates (`lastPMS`, `startDate`, `endDate`, `returnedOn`) are saved as text in the form `YYYY-MM-DD`, because SQLite has no separate date type.
- `returnedOn` stays empty (NULL) until the vehicle is returned. When a vehicle comes back, even before `endDate`, the status changes to RETURNED and the vehicle can be rented again.
- `Rental.renterId` and `Rental.vehicleId` are foreign keys. The app turns on `PRAGMA foreign_keys = ON`, so SQLite refuses a rental that points to a renter or vehicle that does not exist.
- `Vehicle.type` is limited to SEDAN, SUV, or MOTORCYCLE.
- All JDBC statements that use values use `PreparedStatement` with bound parameters.

5. API Integration Design

Public API consumed. FleetRent uses the free NHTSA vPIC (Vehicle Product Information Catalog) REST API from the US National Highway Traffic Safety Administration. It needs no API key. Base address: `https://vpic.nhtsa.dot.gov/api`. We use it so the make and model of every vehicle come from

official data instead of typed text. The catalog lists vehicles sold in the United States, so a few models are not sold in the Philippines (approved by our instructor).

Endpoints called. All are HTTP GET requests that return JSON (`?format=json`). Our three vehicle types map to the NHTSA type words: SEDAN to `car`, SUV to `mpv`, MOTORCYCLE to `motorcycle`.

Endpoint	When we use it
`/vehicles/GetMakesForVehicleType/{type}?format=json`	The user picks a vehicle Type in the Add Vehicle form. It fills the Make list, so a motorcycle make can never be chosen for a Sedan.
`/vehicles/GetModelsForMakeYear/make/{make}[/modelyear/{year}]/vehicleType/{type}?format=json`	The user clicks a make. It fills the Model list. The year part is used only when a Year is chosen (the API has years after 1995). If NHTSA has no model for that year, the app asks again without the year and shows all years.
`/vehicles/GetModelsForMake/{make}?format=json`	The Dashboard question "NHTSA models per make in our fleet (live)", once for each make we own.

Shape of the JSON response. The answer is one object. The part we use is the `Results` array, with one item per model:

```
{
  "Count": 3,
  "Message": "Response returned successfully",
  "Results": [
    { "Make_ID": 448, "Make_Name": "TOYOTA", "Model_ID": 2208, "Model_Name": "Corolla" }
  ]
}
```

The makes answer has the same `Results` array, with `MakeId` and `MakeName` in each item.

How the data is parsed. The class `VpicClient` sends the request with Java's `java.net.http.HttpClient` (10 second connect timeout, 30 second request timeout). `VpicClient.parseModels` reads the JSON text with the Gson library and builds one `VehicleModelInfo` object (make name, model id, model name) for each item in `Results`, skipping items without a model name and sorting by name. `VpicClient.parseMakes` reads `MakeName`, removes duplicates and sorts. Invalid JSON, a missing `Results` list or an HTTP status other than 200 is reported as an error, never a crash.

How the data is used. (1) The parsed objects fill the Make and Model lists of the "Choose from NHTSA" card on the Vehicles tab. The user clicks a make and a model, so the vehicle saved in our SQLite database always has an official make and model that match its type. (2) The Dashboard question "NHTSA models per make in our fleet (live)" calls the API once per make in the fleet and charts how many models NHTSA lists for each. Nothing is cached: every choice makes a new live call. If NHTSA cannot be reached, a red

message says why (no internet, too slow, blocked, server problem), and Make and Model can be typed by hand until it works again.

6. Concurrency Design

Long-running operations. Four operations can take long enough to freeze the window, so each runs on a background thread:

Operation	Where	Why it is slow
NHTSA make and model lists	`VehicleTab.loadMakes`, `VehicleTab.loadModels`	They wait for the internet.
CSV rental import	`ImportTab.runImport`	It reads a file and saves thousands of rows (the demo file has 50,000 rentals).
Car photo search	`VehicleTab.afterAdd`, `VehicleTab.findAllPhotos`	It asks Wikipedia and downloads pictures.
Live dashboard question	`DashboardTab.showLive`	It makes one API call for each make in the fleet.

Threading approach. We use JavaFX `Task` objects. The slow work is inside the Task's `call()` method, and the helper `UiHelper.runInBackground` runs it on its own daemon thread. When it finishes, `setOnSucceeded` or `setOnFailed` runs on the JavaFX Application Thread, so it can safely update labels, lists and charts. During the CSV import, progress is sent to the window with `Platform.runLater`. The background import and the screen share one SQLite connection, so every `Database` method is `synchronized`, and the import saves rentals in batches of 100 inside one transaction.

How the UI stays responsive. The Application Thread only draws the window and handles clicks. While a Task runs, the button that started it is disabled and a `ProgressBar` is shown (an endless bar for API calls, a bar from 0 to 100 percent for the import). The user can switch tabs or type in forms while the work continues. If the work fails, a plain message is shown and the button works again. If the user changes the choice while a request is running, the old answer is ignored when it arrives.

7. Data Mining & Visualization Design

Data analyzed. The Dashboard tab mines the rentals saved in our SQLite database (entered by hand or imported from CSV) together with the vehicle table. Only RETURNED rentals are counted, because only they have a final cost. The class `FleetAnalytics` does the calculations and has no screen code. The user picks a question and a From and To date.

Aggregation and frequency analysis. Every number is calculated from the database each time; nothing is hard-coded.

Question	Mining step
Most rented vehicle type	Frequency count of returned rentals per type (Sedan, SUV, Motorcycle).
Fleet utilization per month (%)	For each month: rented vehicle-days divided by (vehicles x days in the month), times 100.
Revenue by vehicle type	Sum of each rental's cost (`computeRentalCost` of its vehicle) per type.
Rentals per month	Frequency count of rentals per year and month.
Top 5 rented models	Frequency count per make and model, sorted, cut to 5.
NHTSA models per make (live)	Number of models NHTSA lists for each make in our fleet.

A summary line reports the number of rentals, total revenue, best earning type, busiest month and average rental length. A recursive method, `CategoryNode.totalMileage`, also totals the mileage of the category tree (Fleet > Car > Sedan) and is shown under the chart.

Charts and why we chose them. We use two JavaFX charts. A `BarChart` is used for questions that compare separate groups (rentals per type, revenue per type, top models, models per make), because bars make the difference between groups easy to compare. A `LineChart` is used for the two questions that change over time (utilization per month, rentals per month), because a line shows the rise and fall from one month to the next. Both charts have a title, labeled axes and one series, and they redraw whenever the question or the dates change. We do not use a pie chart because 12 months or many models would be hard to read.

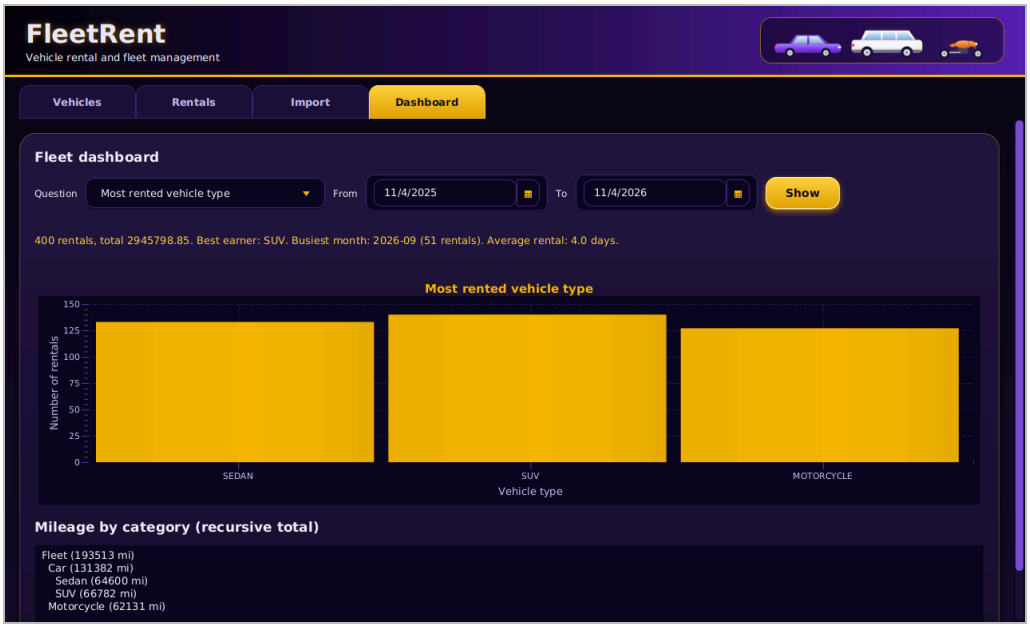


Figure 4: Dashboard bar chart: most rented vehicle type (sample data)



Figure 5: Dashboard line chart: fleet utilization per month (sample data)

8. Testing Summary

The project has 118 JUnit 5 tests in 17 test classes. They run with `mvn test` and all pass (0 failures, 0 errors). Database tests use an in-memory SQLite database, and API tests use a small fake web server on the same computer, so no test needs the internet.

Area	Test classes	Tests
Vehicle classes, category tree, Queue and Stack, vehicle types	`VehicleCostTest`, `CategoryNodeTest`, `QueueStackTest`, `VehicleTypesTest`	15
Rental rules	`RentalMathTest`, `RentalServiceTest`	13
Database and startup	`DatabaseTest`, `MainTest`	7
CSV import and demo data	`RentalCsvImporterTest`, `SampleDataTest`	11
NHTSA and Wikipedia API code	`VpicClientTest`, `CarImageClientTest`, `CarImageStoreTest`	34
Data mining	`FleetAnalyticsTest`	18
Input rules, error messages, colors	`ValidatorsTest`, `ErrorMessagesTest`, `ColorNamesTest`	20

9. Individual Contributions

To be completed in Week 11.

Appendix

To be completed in Week 11.